What Does 99% Accuracy Measure? A Reproducible Audit of Shortcut Learning in a Widely Used Fake News Corpus

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***Abstract*—Text classifiers trained on the ISOT/KAGGLE "Fake and Real News" corpus routinely report accuracy and F_1 above 0.98, a level of performance that sits uneasily beside the difficulty of assessing veracity. We ask what those numbers actually measure. Treating a conventional TF–IDF and linear-classifier pipeline as a measurement instrument rather than a contribution, we audit the corpus along three leakage channels and two distribution-shift protocols, with all code, seeds, and derived numbers released. Three findings emerge. First, the benchmark is partly degenerate: a classifier that discards the article text entirely and observes only the `subject` metadata field attains $F_1 = 1.000$, because the topic labels of the two classes do not overlap. Second, removing all three leakage channels—metadata, a newswire source tag present in 99.2% of real articles, and 6,251 duplicate documents that contaminate 19.4% of a naive test split—lowers F_1 by only 1.21 points (0.9935 to 0.9814), because the residual signal is diffuse editorial style rather than a small set of giveaway tokens: nine representation variants span just 0.6 F_1 points, 543 labelled documents already reach $F_1 = 0.933$, and deleting the 1,000 highest-weight unigrams still leaves 0.926. Third, that style signal does not transfer. Under a topic-disjoint protocol, average precision falls from 0.9995 to 0.9475 and deployed F_1 from 0.9905 to 0.8067; a prior-matched resampling analysis attributes most of the latter collapse to class-prior shift and threshold miscalibration ($F_1 = 0.9427$) while confirming a genuine 5.2-point loss of discrimination. Temporal transfer within the corpus is by contrast nearly lossless. A fine-tuned DistilBERT sharpens rather than resolves the problem: it is stronger in-distribution ($F_1 = 0.9993$) yet degrades far more under topic shift, losing 12.9 points of average precision against the linear model's 5.2 and reaching prior-matched F_1 of only 0.689, while its in-domain validation F_1 of 0.9995 gives no warning of that collapse. We conclude that within-corpus scores on this benchmark quantify source and topic separability rather than veracity assessment, that added capacity exploits the shortcut more efficiently instead of avoiding it, and we recommend metadata-only, small-sample, and topic-disjoint baselines as inexpensive diagnostics that any future study on this corpus can report.**

***Index Terms*—fake news detection, shortcut learning, dataset bias, data leakage, distribution shift, text classification, TF–IDF, reproducibility, evaluation methodology**

I. Introduction

False news propagates measurably faster and more widely than true news on social platforms [1], and its documented reach during the 2016 United States election [2] motivated sustained interest in automated detection. The task, however, is difficult in a way that resists purely lexical solutions: deciding whether a claim is true generally requires evidence external to the text [3], [4]. It is therefore notable that a large body of applied work reports near-perfect performance on the task, with accuracies above 0.99 on public corpora [5]–[7]. Either the problem is easier than its formulation suggests, or the reported numbers measure something other than veracity assessment. This paper investigates the second possibility for one specific, heavily used corpus.

Our object of study is the "Fake and Real News" dataset [8], a redistribution of the corpus introduced by Ahmed et al. [5], [6] and among the most frequently used public resources for this task [7]. It pairs 21,417 genuine articles drawn from a single newswire against 23,481 articles from outlets flagged by fact-checking organisations. That construction is convenient and, as we show, consequential: the two classes differ systematically in provenance and house style, not only in truthfulness.

The phenomenon we investigate is *shortcut learning*: a model attaining high benchmark scores by exploiting surface regularities that correlate with the label in the dataset but not in the underlying task [9]. Shortcut learning is well documented in natural language inference, where hypothesis-only baselines and annotation artifacts account for much of apparent model competence [10]–[13], and in computer vision, where dataset identity is itself predictable [14]. For fake news specifically, Bozarth and Budak [15] document how evaluation choices drive reported performance, and Schuster et al. [16] show that stylometric cues fail when provenance and veracity are decoupled. What has been missing, to the best of our knowledge, is a per-channel, fully reproducible decomposition for this particular corpus, combined with a shift analysis that separates calibration effects from genuine degradation.

We deliberately do not propose a new architecture. A stronger model would confound the question: if the benchmark is degenerate, a higher score is evidence about the benchmark, not the model. We instead fix a transparent, fully interpretable pipeline—TF–IDF features with four linear classifiers—and vary the *data* and the *evaluation protocol*, using the classifier as a probe. Linear models make this strategy viable because every decision decomposes into per-token weights that can be inspected and ablated directly.

Contributions.

1) A reproducible audit protocol (Section IV) that isolates three leakage channels in the ISOT/KAGGLE corpus and quantifies each independently, released as executable code with fixed seeds (Section VI).
2) Evidence that the benchmark is partly degenerate: a metadata-only classifier that never sees the article text attains $F_1 = 1.000$ (Section VII-B).
3) A demonstration that the residual post-mitigation signal is *diffuse* rather than localised, established through representation ablations, a learning curve, and a top-K feature-deletion probe (Section VII-C).
4) A prior-controlled distribution-shift analysis (Section VII-D) that decomposes an 18.4-point deployed F_1 collapse into a class-prior/calibration component and a genuine 5.2-point loss of average precision.
5) A capacity control (Section VII-E) in which a fine-tuned DistilBERT, trained on identical documents and splits, is stronger in-distribution but loses 2.5 times more average precision under topic shift—evidence that the limitation is a property of the data rather than of model expressiveness.
6) Concrete, low-cost diagnostics we recommend as standard practice for future work on this corpus (Section VIII).

II. Related Work

A. Fake News Detection and Its Benchmarks

Surveys of the area distinguish content-based approaches from those exploiting social context and external evidence [4], [17], [18]. Purely content-based detection is attractive operationally—it needs only the article—but is theoretically limited, since textual form underdetermines factual accuracy. Within that paradigm, Pérez-Rosas et al. [19] report linguistic-feature classifiers across several domains, while Horne and Adalı [20] find that headlines alone carry substantial discriminative signal—an observation our input-field ablation revisits in Section VII-C. Zellers et al. [21] extend the problem to machine-generated articles, where provenance and veracity come apart by construction. Benchmarks reflect this tension. LIAR [22] supplies short PolitiFact-labelled claims with fine-grained veracity labels; FEVER [3] pairs claims with Wikipedia evidence, making the evidence-retrieval step explicit; NELA-GT [23] provides large multi-labelled article collections with source-level annotations. The ISOT/KAGGLE corpus [5], [6], [8] differs in an important respect: its two classes were collected from disjoint sets of publishers. Labels therefore carry provenance information, a property central to our analysis.

Reported results on the corpus are uniformly high. Ahmed et al. obtained accuracies in the high nineties using n-gram features with linear models [5], [6]; the benchmark study of Khan et al. [7] finds that simple models remain competitive with neural alternatives across several fake-news corpora, including this one, and that scores cluster near the ceiling. Transformer-based systems such as FakeBERT [24] report comparable figures. Our reading of this pattern is that near-ceiling agreement across model families of very different capacity is itself diagnostic: when a linear bag-of-words model and a pretrained transformer perform indistinguishably, the discriminative information is likely to be shallow.

B. Shortcut Learning and Dataset Artifacts

Geirhos et al. [9] formalise shortcut learning as reliance on decision rules that succeed on benchmark data but fail under distribution shift. In NLI, Gururangan et al. [10] and Poliak et al. [11] show that hypothesis-only models substantially outperform chance, revealing annotation artifacts; McCoy et al. [12] and Niven and Kao [13] show that apparently competent models rely on syntactic heuristics and spurious statistical cues. Diagnostic methodologies developed in response include contrast sets [25] and behavioural testing [26]. Our metadata-only and small-sample probes are direct analogues of the hypothesis-only baseline: cheap experiments whose success indicates that the benchmark, not the model, deserves scrutiny.

C. Leakage and Evaluation Protocol

Kaufman et al. [27] give the canonical taxonomy of leakage; Kapoor and Narayanan [28] document its prevalence and its role in irreproducible machine-learning claims across scientific fields. Duplicate records spanning train and test are a recognised instance. Gorman and Bedrick [29] show that conclusions drawn from a single standard split can reverse under resampling, motivating our use of cross-validation and bootstrap intervals alongside a fixed split. On protocol design specifically, Bozarth and Budak [15] show that performance rankings for fake-news classifiers depend strongly on evaluation choices, and argue for source-aware splits. Schuster et al. [16] demonstrate that stylometric detectors degrade sharply once provenance is decoupled from veracity. Our topic-disjoint protocol operationalises that concern for this corpus, and our prior-matched analysis adds a component their setting does not isolate: how much of an observed collapse is attributable to class-prior shift rather than to loss of discriminative power.

D. Positioning

Relative to prior work, this paper contributes neither a new model nor a new dataset. Its contribution is measurement: a per-channel quantification of leakage in a specific widely used corpus, an explicit demonstration that the surviving signal is distributed rather than concentrated, and a shift analysis that separates two mechanisms usually reported as one number. Every figure in the paper is regenerated by a released script from the raw corpus.

III. Problem Statement

Let $\mathcal{D} = \{(d_i, y_i)\}_{i=1}^{N}$ be a corpus of news documents $d_i \in \mathcal{X}$ with labels $y_i \in \{0,1\}$, where $y=1$ denotes *fake*. The nominal task is to learn $f : \mathcal{X} \to \{0,1\}$ minimising expected risk under the deployment distribution $\mathcal{P}_{\text{dep}}$:

$$R_{\text{dep}}(f) \;=\; \mathbb{E}_{(d,y)\sim\mathcal{P}_{\text{dep}}}\big[\mathbb{I}[f(d) \neq y]\big]. \tag{1}$$

Benchmark practice instead reports empirical risk on a held-out split of the same corpus, i.e. an estimate of $R_{\text{bench}}(f)$ under the corpus distribution $\mathcal{P}_{\text{bench}}$. The two coincide only if $\mathcal{P}_{\text{bench}} \approx \mathcal{P}_{\text{dep}}$.

We formalise the failure mode of interest as follows. Write each document as a pair $d = (s, c)$ where c is veracity-bearing content and s is a *provenance signature*: stylistic, formatting, and editorial regularities determined by the publishing organisation. In $\mathcal{P}_{\text{bench}}$ the two classes were sampled from disjoint publisher sets, so s and y are strongly dependent:

$$I(s;y)\Big|_{\mathcal{P}_{\text{bench}}} \;\gg\; I(s;y)\Big|_{\mathcal{P}_{\text{dep}}}, \tag{2}$$

where $I(\cdot;\cdot)$ denotes mutual information. A learner minimising empirical risk has no incentive to prefer c over s; if s is more easily extracted, the learned rule will exploit it. Benchmark risk is then a biased estimate of deployment risk, and the bias is not removed by increasing model capacity or dataset size—only by changing the data or the protocol.

This yields three empirical questions, which organise our experiments:

- Q1 *Explicit leakage.* How much of the benchmark score is attributable to identifiable artifacts—metadata fields, source tags, and duplicate documents—that can be removed by preprocessing?
- Q2 *Residual signal structure.* After removing those artifacts, is the remaining signal concentrated in a few tokens (hence removable) or diffusely distributed across editorial style (hence not)?
- Q3 *Transfer.* Does the surviving decision rule remain valid when the topic mix or time period changes, and how much of any observed degradation reflects loss of discrimination as opposed to class-prior shift?
- Q4 *Capacity.* Is the limitation a property of the instrument or of the data? If a pretrained contextual model, which can represent veracity-bearing content far better than a bag of words, nonetheless degrades at least as much under shift, the constraint lies in $\mathcal{P}_{\text{bench}}$ rather than in model expressiveness.

IV. Proposed Methodology

A. Design Rationale

The instrument must be transparent, cheap, and deterministic, since it is applied across 20 experimental conditions. We therefore use sparse TF–IDF features with linear classifiers. This choice is deliberate on three grounds. First, interpretability: a linear model assigns one scalar weight per token, so the top-K deletion probe of Section VII-C has a well-defined meaning [30]. Second, determinism: with a fixed seed and a coordinate-descent solver the pipeline is bit-reproducible, which matters when differences of interest are on the order of 10^{-2}. Third, capacity control: if a low-capacity model already saturates the benchmark, high scores cannot be attributed to sophisticated inference. The trade-off—no modelling of word order beyond short n-grams, no lexical generalisation—is accepted knowingly, and Section IX states where it constrains our conclusions.

Distributed word representations [31], [32], subword linear models [33], character-level convolutional networks [34], and pretrained transformers [35], [36] are all defensible alternatives, and we return to them in Section X. None is preferable *as an instrument* for the present question, because each replaces inspectable per-token weights with distributed parameters that the deletion probe of Section VII-C could not target. Sparse lexical features with linear models also remain strong baselines for topical text classification [37], [38], so little discriminative power is sacrificed.

B. Corpus Construction and Leakage Control

Algorithm 1 specifies corpus construction. Three controls are applied, each independently switchable so that its contribution can be measured:

(C1) Metadata exclusion. The `subject` and `date` fields are discarded. In this corpus the real class carries only the subjects `politicsNews` and `worldnews` while the fake class carries six entirely different values (Table I); the supports are disjoint, so `subject` alone determines the label. The `date` fields are also not exchangeable: 2,926 fake articles fall outside the date range spanned by the real class.

(C2) Source-tag removal. Real articles are newswire copy that typically opens with a dateline of the form `CITY (Reuters) --` ; the agency name appears in 99.21% of real articles and 0.04% of fake ones. We strip the leading dateline and every remaining occurrence of the agency name. The transformation is applied symmetrically to both classes, so it introduces no new asymmetry.

(C3) De-duplication. Exact duplicates of the cleaned document string are collapsed to a single instance *before* splitting. Under a naive protocol that omits this step, 19.37% of test documents also occur verbatim in the training set (Section VII-B), so test performance partly measures memorisation.

Remaining normalisation is deliberately conservative and purely regular: the title and body are concatenated, case is folded, URLs and e-mail addresses are removed, characters outside `[a--z']` are mapped to whitespace, runs of whitespace are collapsed, and documents shorter than 20 characters are dropped. We do not stem or lemmatise. Stemming would add a dependency on an external resource without addressing any of Q1–Q3, and with bigrams and stop-word removal already in place its effect on this corpus is marginal.

C. Feature Representation

Documents are mapped to sparse TF–IDF vectors [39], [40] over word unigrams and bigrams. With N training documents

Algorithm 1 Leakage-controlled corpus construction

Require: Raw tables $T_{\text{real}}, T_{\text{fake}}$; flags $\textsc{c}1, \textsc{c}2, \textsc{c}3$; seed σ
Ensure: Disjoint splits $\mathcal{S}_{\text{tr}}, \mathcal{S}_{\text{va}}, \mathcal{S}_{\text{te}}$

$\mathcal{D} \leftarrow \text{concat}(T_{\text{real}} \times \{0\},\ T_{\text{fake}} \times \{1\})$
for all $(d, y) \in \mathcal{D}$ **do**
  $u \leftarrow \textsc{title}(d) \| \text{`` ''} \| \textsc{body}(d)$
  if $\neg\textsc{c}1$ **then** ▷ leakage condition only
    $u \leftarrow \textsc{subject}(d) \| \text{`` ''} \| u$
  end if
  $u \leftarrow \textsc{lowercase}(u)$
  if $\textsc{c}2$ **then**
    $u \leftarrow \textsc{stripDateline}(u)$; $u \leftarrow \textsc{removeAgencyName}(u)$
  end if
  $u \leftarrow \textsc{removeUrlsEmails}(u)$; $u \leftarrow \textsc{keepAlpha}(u)$
  $u \leftarrow \textsc{collapseSpace}(u)$
end for
$\mathcal{D} \leftarrow \{(u, y) \in \mathcal{D} : |u| \geq 20\}$
if $\textsc{c}3$ **then**
  $\mathcal{D} \leftarrow \textsc{dropDuplicates}(\mathcal{D}, \text{key} = u)$
end if
$(\mathcal{D}', \mathcal{S}_{\text{te}}) \leftarrow \textsc{stratifiedSplit}(\mathcal{D}, 0.20, \sigma)$
$(\mathcal{S}_{\text{tr}}, \mathcal{S}_{\text{va}}) \leftarrow \textsc{stratifiedSplit}(\mathcal{D}', 0.125, \sigma)$ ▷ $0.10/0.80$
return $\mathcal{S}_{\text{tr}}, \mathcal{S}_{\text{va}}, \mathcal{S}_{\text{te}}$

and $\text{df}(t)$ the document frequency of term t, we use sublinear term frequency and smoothed inverse document frequency,

$$\text{tf}'(t, d) = 1 + \ln \text{tf}(t, d), \tag{3}$$

$$\text{idf}(t) = \ln \frac{1 + N}{1 + \text{df}(t)} + 1, \tag{4}$$

followed by ℓ_2 normalisation of each document vector,

$$x_{d,t} = \frac{\text{tf}'(t, d)\, \text{idf}(t)}{\left\| \text{tf}'(\cdot, d) \odot \text{idf}(\cdot) \right\|_2}. \tag{5}$$

Sublinear scaling prevents a term repeated many times in one document from dominating; ℓ_2 normalisation removes document-length effects, which matters here because the classes differ in mean length (436.7 versus 395.8 tokens, Table I). Terms occurring in fewer than five documents or more than 90% of documents are discarded, English stop words are removed, and the vocabulary is capped at $50{,}000$ features by corpus frequency.

Critically, the vectoriser is fitted on the training split only and applied unchanged to validation and test. Fitting on the full corpus would leak document-frequency statistics of the evaluation data into the representation—a subtle but real instance of the leakage this paper studies.

D. Classifiers

Four linear classifiers are trained on identical features. All four are standard; we summarise them to fix notation and to make the interpretability argument precise. Let $x \in \mathbb{R}^V$ be a document vector, $w \in \mathbb{R}^V$ a weight vector, and b a bias.

Logistic regression models $P(y{=}1 \mid x) = \sigma(w^\top x + b)$ with $\sigma(z) = (1 + e^{-z})^{-1}$, fitted by ℓ_2-regularised maximum likelihood. With labels $\tilde{y} \in \{-1, +1\}$ the objective solved by the coordinate-descent solver [41] is

$$\min_{w,b}\ \tfrac{1}{2}\|w\|_2^2 + C \sum_{i=1}^{n} \ln\bigl(1 + e^{-\tilde{y}_i (w^\top x_i + b)}\bigr). \tag{6}$$

It is our reference probe because w is directly interpretable and because $\sigma(\cdot)$ yields calibrated scores.

Multinomial naive Bayes [42] estimates per-class term distributions with additive smoothing α,

$$\begin{aligned} \hat{P}(t \mid y) &= \frac{N_{y,t} + \alpha}{N_y + \alpha |V|}, \\ \hat{y} &= \arg\max_y \Bigl[\ln \hat{P}(y) + \sum_t x_t \ln \hat{P}(t \mid y)\Bigr], \end{aligned} \tag{7}$$

under the assumption that terms are conditionally independent given the class. The assumption is violated by construction in text—overlapping unigrams and bigrams are strongly dependent—and we use the model as a deliberately misspecified reference point [43], [44].

Linear support vector classification [37], [45] maximises the margin under squared hinge loss,

$$\min_{w,b}\ \tfrac{1}{2}\|w\|_2^2 + C \sum_{i=1}^{n} \max\bigl(0,\, 1 - \tilde{y}_i (w^\top x_i + b)\bigr)^2. \tag{8}$$

The passive–aggressive classifier [46] is an online large-margin learner. On example $(x_t, \tilde{y}_t)$ with hinge loss $\ell_t = \max(0, 1 - \tilde{y}_t\, w_t^\top x_t)$, the PA-I update is

$$\tau_t = \min\Bigl(C,\ \frac{\ell_t}{\|x_t\|_2^2}\Bigr), \qquad w_{t+1} = w_t + \tau_t\, \tilde{y}_t\, x_t, \tag{9}$$

i.e. the weights are left unchanged when the margin is already satisfied ($\ell_t = 0$) and otherwise moved by the smallest step that corrects the current example, with C bounding the step. In all four models C is an inverse regularisation strength: larger C penalises weight magnitude less.

E. Evaluation Protocols

Deliberately shifting the test distribution is standard practice for exposing brittle decision rules [47], [48]. Three protocols partition the same cleaned corpus.

Random (standard). Stratified $70/10/20$ train/validation/test. This reproduces conventional practice and provides the reference score.

Topic-disjoint. Training and test pools are drawn from disjoint `subject` values: `politicsNews` (real) and `News`, `politics`, `left-news` (fake) for training; `worldnews` (real) and `Government News`, `US_News`, `Middle-east` (fake) for test. Validation is carved from the training pool only, so no target-domain labels are available at model-selection time—the realistic deployment condition. Because subject values also proxy for topic and sub-publication, this protocol perturbs provenance and topic jointly.

Temporal. Documents are ordered by publication date and split chronologically $70/10/20$, restricted to the window in

Algorithm 2 Prior-controlled shift evaluation

Require: Fitted model f with scores $s(\cdot)$; test set $\mathcal{S}_{\text{te}}$; reference prior $\pi^\star$; repeats M; seed σ

$\hat{y} \leftarrow f(\mathcal{S}_{\text{te}});\;\; s \leftarrow s(\mathcal{S}_{\text{te}})$
$F_1^{\text{def}} \leftarrow F_1(y, \hat{y})$ ▷ deployed operating point
$F_1^{\text{orc}} \leftarrow \max_\theta F_1\big(y, \mathbb{I}[s > \theta]\big)$ ▷ isolates calibration
$\text{AP} \leftarrow \textsc{averagePrecision}(y, s)$ ▷ threshold-free
$\mathcal{P} \leftarrow \{i : y_i{=}1\};\;\; \mathcal{N} \leftarrow \{i : y_i{=}0\}$
$m \leftarrow \text{round}\big(|\mathcal{P}|(1-\pi^\star)/\pi^\star\big)$
for $j \leftarrow 1$ **to** M **do**
  $\mathcal{N}_j \sim \textsc{sampleWithoutReplacement}(\mathcal{N}, m; \sigma)$
  $\phi_j \leftarrow F_1\big(y_{\mathcal{P}\cup\mathcal{N}_j}, \hat{y}_{\mathcal{P}\cup\mathcal{N}_j}\big)$
end for
return F_1^{def}, F_1^{orc}, AP, $\text{mean}(\phi)$, $\text{sd}(\phi)$

which both classes are present so that the test period is not trivially single-class.

Both shift protocols alter the class prior as a side effect (Table II). Since F_1 depends on the prior, a raw F_1 comparison across protocols conflates prior shift with loss of discrimination. Algorithm 2 therefore reports four complementary quantities: F_1 at the default threshold (what a deployed system would achieve), threshold-oracle F_1 (the best any cutoff on the model's own scores could achieve, isolating miscalibration), average precision (threshold-free ranking quality), and F_1 on prior-matched subsamples in which negatives are resampled so that the fake ratio equals that of the random split. Only the last two support a like-for-like comparison of discriminative power.

F. Capacity Control: Fine-Tuned DistilBERT

Answering Q4 requires a model of substantially greater capacity evaluated under identical conditions. We fine-tune DistilBERT [49], a six-layer distillation of BERT [36] with roughly 66M parameters, whose subword tokenisation and contextual self-attention [35] can in principle represent the semantic content that a bag of words cannot.

Comparability is the whole point of the experiment, so two things are held fixed and one is deliberately allowed to differ. The document set and the split assignment are *identical*: we rebuild the primary frame preserving the original row index, replicate each protocol exactly, and assert that the resulting split cardinalities match those in Table II before training proceeds. The leakage controls C1–C3 are also identical.

What differs is surface normalisation. The linear pipeline maps every character outside `[a--z']` to whitespace, which suits a bag-of-words model but discards punctuation and casing that a subword transformer can exploit. Applying that destructive normalisation would handicap the transformer for reasons unrelated to Q4, so the transformer receives a minimally normalised variant of the *same* documents: the source tag, URLs, and e-mail addresses are removed and whitespace is collapsed, but punctuation and digits are retained. This asymmetry favours the transformer, which is the conservative direction for our argument—we are testing whether extra capacity *helps*, so giving it the better input strengthens any negative result.

Training uses the recipe standard for this model class: batch size 16, learning rate 2×10^{-5} with linear warmup over the first 10% of steps, two epochs, mixed-precision arithmetic, and a maximum sequence length of 256 subword tokens. No hyperparameter search was performed, for the same reason it was omitted for the linear models. Model selection follows the same rule as the linear models—the per-epoch checkpoint with the best *validation* F_1 is the one evaluated on test. Because the corpus mean of 415 words exceeds 256 subword tokens, we additionally repeat the random protocol at a 512-token limit to confirm that truncation is not what determines the outcome.

G. Metrics

We report accuracy, and precision, recall, and F_1 for the positive (fake) class:

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}}, \quad R = \frac{\text{TP}}{\text{TP}+\text{FN}}, \quad F_1 = \frac{2PR}{P+R}. \qquad (10)$$

Taking *fake* as positive makes the error types operationally meaningful: a false positive suppresses a legitimate article, a false negative lets misinformation through. We additionally report ROC-AUC [50], which equals the probability that a randomly chosen fake article receives a higher score than a randomly chosen real one and is invariant to the decision threshold, and average precision, which is more informative than ROC-AUC when the positive class is rare—as it becomes under both shift protocols. Model selection uses validation F_1 throughout, a rule fixed before any test set was examined.

V. System Architecture

Figure 1 shows the system. It is organised so that exactly one factor varies per experimental condition: the preprocessing controls C1–C3 of Section IV-B, the representation hyperparameters of Equations (3)–(5), or the splitting protocol of Section IV-E. Configuration is centralised in a single module and serialised into the results file with every run, so each reported number carries the exact settings that produced it.

A. Computational Complexity

Let N be the number of training documents, $\bar{L}$ the mean token count, V the vocabulary size, and z the mean number of non-zero features per document. Vectorisation is $O(N\bar{L})$ time, dominated by tokenisation and n-gram enumeration, with a vocabulary-construction pass over $O(N\bar{L})$ candidate terms before frequency pruning. Training the linear models costs $O(\kappa N z)$ for κ passes: coordinate descent for logistic regression and the SVM [41], a single closed-form counting pass for naive Bayes, and κ online epochs for passive–aggressive. Inference is $O(z)$ per document—one sparse dot product.

The measured quantities corroborate these rates and explain why the audit is cheap enough to repeat across 20 conditions. On the training split $z = 190.1$ of $V = 50{,}000$ features, i.e. a density of 0.38%; the sparse matrix holds 5.17×10^6 non-zeros in $\approx 62\,\text{MB}$ at 12 bytes per stored value, whereas a dense

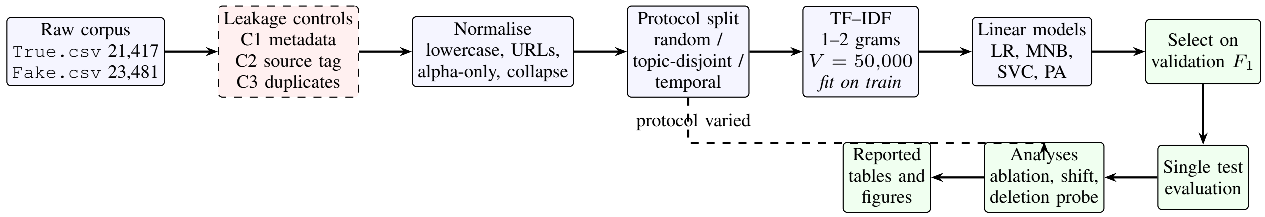

Fig. 1. Audit architecture. The measurement pipeline (solid, blue) is held fixed while the leakage controls C1–C3 (dashed, red) and the evaluation protocol are varied. The validation split governs model selection; the test split is evaluated exactly once per condition. Every reported figure is regenerated from the raw corpus by the released scripts.

equivalent would require ≈ 10.9 GB. Sparse storage is thus not an optimisation but a precondition. Section VII-F reports wall-clock figures.

VI. Experimental Setup

Data. The corpus is obtained from its public distribution [8] and is not redistributed with our code. Table I reports its composition together with the leakage indicators discussed in Section IV-B; all values are computed by the released script rather than quoted from prior work.

Hardware and software. All experiments run on a single machine: Intel Core i7-14700HX (28 logical cores), 15.7 GB RAM, Windows 10 (AMD64). The linear pipeline uses the CPU only; the DistilBERT control uses one NVIDIA GeForce RTX 4050 Laptop GPU (6 GB), peaking at 3.9 GB of device memory. The stack is Python 3.10.11 with scikit-learn 1.7.2 [51], NumPy 1.24.3 [52], SciPy 1.15.3 [53], pandas 2.3.3 [54], and Matplotlib 3.10.8 [55]; the transformer control adds PyTorch 2.6.0 (CUDA 12.4) and Transformers 4.56.1. Exact versions are pinned in the released `requirements.txt`.

Reproducibility. A single seed ($\sigma = 42$) governs the splits and all stochastic model components. Text normalisation is purely regular; the vocabulary is frequency-ordered and deterministic; logistic regression uses the deterministic `liblinear` solver. Two independent executions of the released pipeline produced bit-identical metrics. This guarantee covers the linear pipeline; the DistilBERT control is seeded identically but runs mixed-precision CUDA kernels whose reduction order is not guaranteed, so its figures should be expected to reproduce to roughly three decimal places rather than exactly. Following community recommendations on reporting practice [56], [57] we release code, seeds, configuration snapshots, and the machine-readable results file from which every table and figure in this paper is generated. Appendix D contains the reproducibility checklist and Appendix A the full hyperparameter listing.

Statistical procedure. Point estimates on the fixed test split are accompanied by percentile bootstrap 95% confidence intervals over 2000 resamples of the test set [58]. Because conclusions from a single split can be fragile [29], we additionally report five-fold stratified cross-validation F_1 on the union of the training and validation splits, refitting the vectoriser within each fold. Pairwise model comparisons use McNemar's exact test on discordant predictions [59], [60], the appropriate paired test when two classifiers are evaluated on the same test set [61]; omnibus procedures designed for comparisons across many datasets [62] do not apply to a single-corpus design. Prior-matched F_1 is averaged over $M = 20$ resamples (Algorithm 2).

TABLE I
Corpus composition and leakage indicators. All quantities are computed from the raw distribution by the released script.

Property	Real (y=0)	Fake (y=1)
Articles (raw)	21,417	23,481
`subject` values (disjoint supports)		
`politicsNews` / `worldnews`	11,272 / 10,145	–
`News` / `politics`	–	9,050 / 6,841
`left-news` / `Government News`	–	4,459 / 1,570
`US_News` / `Middle-east`	–	783 / 778
Contains agency name "(Reuters)"	99.21%	0.04%
Duplicate body strings	225	6,026
Bodies shorter than 50 characters	1	835
Mean tokens per document (cleaned)	395.8	436.7
After normalisation and de-duplication		
Documents retained	38,826 of 44,898	
Fake proportion	0.461	
Date span	2015-03-31 to 2018-02-19	

TABLE II
Evaluation protocols. Both shift protocols alter the class prior as a side effect, which motivates the prior-controlled analysis of Algorithm 2.

Protocol	Train	Val.	Test	Test π_{fake}
Random (standard)	27,177	3,883	7,766	0.461
Topic-disjoint	24,326	3,476	11,024	0.119
Temporal	25,758	3,679	7,361	0.113

Sanity control. As a check that the harness itself introduces no leakage, we ran the complete primary pipeline with the labels randomly permuted. Test ROC-AUC was 0.4974, statistically indistinguishable from the chance value of 0.5, confirming that the reported performance originates in the data rather than in an implementation defect.

VII. Results

A. Reference Performance

Table III reports the four classifiers under the standard random protocol. Passive–aggressive attains the best validation F_1 (0.9924) and is selected; on the test split it reaches accuracy

TABLE III
REFERENCE PERFORMANCE UNDER THE RANDOM PROTOCOL. SELECTION USES VALIDATION F_1; TEST COLUMNS ARE COMPUTED ONCE FOR THE SELECTED MODEL AND, FOR COMPLETENESS, FOR THE REMAINING MODELS. CV IS FIVE-FOLD STRATIFIED F_1 ON TRAIN ∪ VALIDATION. THE FINAL COLUMN GIVES MCNEMAR p-VALUES AGAINST THE SELECTED MODEL.

	Val.	Test					
Model	F_1	Acc.	P	R	F_1	AUC	CV F_1
Logistic regression	0.9840	0.9830	0.9906	0.9723	0.9814	0.9987	0.9829±.0030
Multinomial NB	0.9569	0.9592	0.9538	0.9578	0.9558	0.9911	0.9570±.0010
LinearSVC	0.9922	0.9909	0.9933	0.9869	0.9901	0.9995	0.9906±.0015
Passive–aggressive†	**0.9924**	**0.9912**	**0.9933**	**0.9877**	**0.9905**	**0.9995**	**0.9916**±.0015

†selected. McNemar vs. selected: LinearSVC p=0.65 (n.s.); logistic regression p=3.2×10^{-12}; naive Bayes p=3.5×10^{-55}. Bootstrap 95% CI (selected): Acc. [.9892, .9933], F_1 [.9883, .9928], P [.9905, .9958], R [.9839, .9913], AUC [.9992, .9997].

0.9912, precision 0.9933, recall 0.9877, F_1 0.9905, and ROC-AUC 0.9995, reproducing the range reported in the literature for this corpus [6], [7]. Its confusion matrix (Figure 2) contains 24 false positives and 44 false negatives out of 7,766 test documents. Bootstrap intervals are tight—F_1 0.9905, 95% CI $[0.9883, 0.9928]$—and five-fold cross-validation agrees closely (0.9916 ± 0.0015), so the reference figure is not an artefact of the particular split.

Two observations already qualify the headline number. First, the three margin- and likelihood-based models are separated by very little: McNemar's exact test finds no significant difference between passive–aggressive and LinearSVC (n_{01}=8, n_{10}=11, $p = 0.65$), so the selected "winner" is statistically indistinguishable from the runner-up, and reporting it as the best model would overstate what the data support. Differences against logistic regression ($p = 3.2 \times 10^{-12}$) and naive Bayes ($p = 3.5 \times 10^{-55}$) are significant. Second, the mis-specified naive Bayes baseline still reaches F_1 0.9558. When a model whose independence assumption is violated by construction lands within four points of the best system, the task as posed is unlikely to require sophisticated inference.

B. Q1: Explicit Leakage Channels

Table IV and Figure 3 isolate the three channels using logistic regression as a fixed probe.

The most consequential result requires no classifier of the article at all. A model whose only input is the `subject` field—the article text discarded entirely—attains $F_1 = 1.000$ and accuracy $= 1.000$ on a held-out split, with a vocabulary of nine tokens. This is not a subtle correlation but a deterministic mapping: the two classes have disjoint subject supports (Table I), so any model given this field can recover the label exactly. Any experiment on this corpus that retains `subject` among its features is measuring nothing about text.

Restricting inputs to title and body, the naive configuration reaches F_1 0.9935. De-duplication alone costs 0.70 points and source-tag removal alone 0.49 points; applied together they cost 1.21 points, yielding the primary figure of 0.9814. The duplicate channel has a clear mechanism: under the naive protocol 1,739 of 8,978 test documents (19.37%) occur verbatim in the training set, so roughly a fifth of the test score reflects retrieval of memorised strings rather than generalisation.

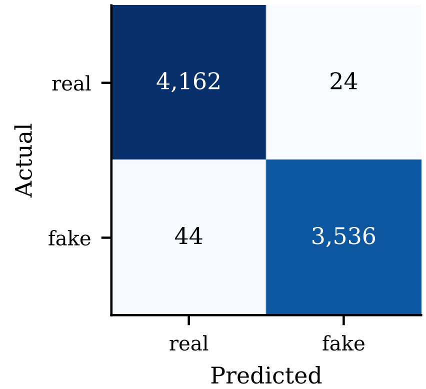

Fig. 2. Confusion matrix of the selected passive–aggressive model on the random-protocol test split ($n = 7{,}766$): 3,536 true positives, 4,162 true negatives, 24 false positives, 44 false negatives. The 24 false positives are legitimate articles that would be suppressed; the 44 false negatives are fake articles that would pass.

TABLE IV
LEAKAGE ABLATION WITH A FIXED LOGISTIC-REGRESSION PROBE. Δ IS THE CHANGE IN TEST F_1 RELATIVE TO THE NAIVE CONFIGURATION. THE METADATA-ONLY ROW USES THE `SUBJECT` FIELD ALONE, WITH THE ARTICLE TEXT DISCARDED.

Configuration	Docs	Acc.	F_1	AUC	ΔF_1
Metadata only (`subject`)	44,898	1.0000	1.0000	1.000	+0.0065
Naive (no mitigation)	44,889	0.9932	0.9935	0.9996	–
+ de-duplication (C3)	38,826	0.9876	0.9865	0.9994	−0.0070
+ source tag removed (C2)	44,889	0.9881	0.9886	0.9992	−0.0049
+ both (primary)	38,826	0.9830	0.9814	0.9987	−0.0121

The honest reading of Table IV is two-sided, and we state both directions. Each channel is real, measurable, and worth removing; a study that ignores all three overstates its result by more than a point and, if it retains metadata, by far more. Yet the mitigations do *not* bring performance down to a plausible level for veracity assessment—0.9814 remains close to the ceiling. Explicit artifacts are therefore not the principal explanation for the benchmark's easiness, which motivates Q2.

C. Q2: The Residual Signal Is Diffuse

Three independent probes indicate that what survives mitigation is not a small set of removable cues but a pervasive stylistic difference.

Representation is nearly irrelevant. Table V varies the feature space over nine configurations spanning n-gram order, vocabulary size across a 35-fold range, term weighting, and stop-word handling. Test F_1 lies between 0.9773 and 0.9835—a spread of 0.62 points. A 5,000-term unigram-and-bigram vocabulary (0.9835) slightly *outperforms* the 50,000-term default and the 175,406-term unrestricted vocabulary (0.9800). When a tenfold reduction in capacity does not degrade performance, the discriminative information is heavily redundant.

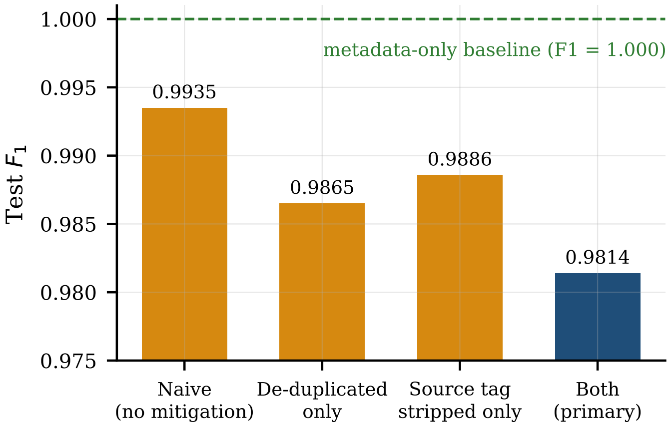

Fig. 3. Effect of the leakage mitigations on test F_1 (logistic-regression probe). Removing all three channels costs 1.21 points, leaving performance close to the ceiling. The dashed line marks the metadata-only baseline, which solves the benchmark exactly without reading any article text.

Very little data suffices. Figure 4 shows the learning curve. With 543 labelled documents—2% of the training split—the probe reaches F_1 0.9334, i.e. 95.1% of the full-data figure; 135 documents already yield 0.6802. Tasks requiring genuine inference do not typically saturate on a few hundred examples.

Deleting the strongest cues is not enough. Ranking unigrams by $|w|$ under the probe and removing the top K from the vocabulary entirely (so bigrams containing them are also destroyed) yields Figure 5: F_1 declines gracefully from 0.9814 at $K = 0$ to 0.9600 at $K = 100$ and 0.9263 at $K = 1000$. Removing the thousand most discriminative unigrams—among them *said*, *video*, *washington*, *image*, *featured*, *getty*, and the weekday names—still leaves a model within 5.5 points of the original. The shortcut is not a token; it is a register.

Figure 6 makes that register legible. The strongest evidence for *real* consists of newswire conventions (*said*, attribution phrasing, weekday datelines, *factbox*); the strongest evidence for *fake* consists of web-publishing conventions (*video*, *featured image*, *getty images*, *watch*). These are properties of editorial workflow and content-management systems, not of factual accuracy—exactly the provenance signature s of Section III.

D. Q3: Transfer Under Distribution Shift

Table VI and Figure 7 report the prior-controlled analysis; Figure 8 contrasts ROC behaviour between protocols.

Under the topic-disjoint protocol the deployed operating point degrades severely: F_1 for the selected model falls from 0.9905 to 0.8067 (−18.4 points), and for logistic regression from 0.9814 to 0.6996 (−28.2 points). Precision drives the collapse (passive–aggressive: 0.9933 → 0.7011) while recall is largely preserved (0.9877 → 0.9498), the signature of a decision threshold that is no longer appropriate for the target class prior, which falls from 0.461 to 0.119.

Decomposing that collapse is essential for an honest claim, and it cuts both ways. Matching the prior by resampling negatives recovers most of the loss (F_1 0.9427 ± 0.0032), and the

TABLE V
REPRESENTATION AND INPUT-FIELD ABLATION (LOGISTIC-REGRESSION PROBE, PRIMARY CLEANING, RANDOM PROTOCOL). PERFORMANCE IS INSENSITIVE TO THE FEATURE SPACE ACROSS A 35-FOLD RANGE OF VOCABULARY SIZE.

Axis	Variant	Vocabulary	Test F_1
n-gram order	unigram	30,473	0.9801
	unigram + bigram (default)	50,000	0.9814
	+ trigram	50,000	0.9820
Vocabulary cap	5,000	5,000	**0.9835**
	10,000	10,000	0.9823
	25,000	25,000	0.9817
	unrestricted	175,406	0.9800
Weighting	without sublinear tf	50,000	0.9773
	stop words retained	50,000	0.9828
Input field	title only	39,615	0.9265
	body only	49,997	0.9772
	title + body (default)	50,000	0.9814

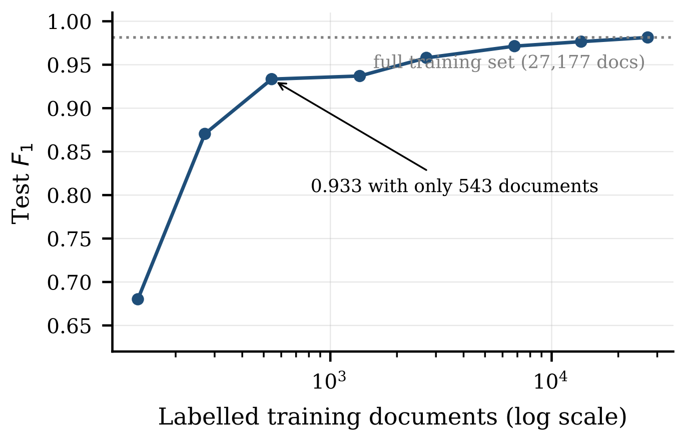

Fig. 4. Learning curve of the logistic-regression probe under the primary configuration. Two per cent of the training data recovers 95.1% of the full-data F_1, indicating a signal that is easy to acquire.

threshold-oracle value (0.8887) shows that a substantial part of the remainder is miscalibration rather than an inability to rank. We therefore do *not* claim an 18-point loss of discriminative power. However, average precision—which is threshold-free and so immune to both effects—falls from 0.9995 to 0.9475, a genuine 5.2-point degradation, and balanced accuracy falls from 0.9910 to 0.9475. Logistic regression degrades further (AP 0.9985 → 0.9018, −9.7 points), indicating that the more heavily regularised probe leans harder on cues that do not transfer. The correct statement is thus: under topic shift the model retains much of its ranking ability but loses a real and measurable fraction of it, and its deployed decision rule becomes unusable without recalibration on target data that a real deployment would not possess.

Temporal shift, by contrast, is nearly benign: average precision moves only from 0.9995 to 0.9964 and prior-matched F_1 is statistically indistinguishable from the random-protocol value (0.9914 ± 0.0013 versus 0.9905). The asymmetry is informative. Publisher house style is stable over the 2016–

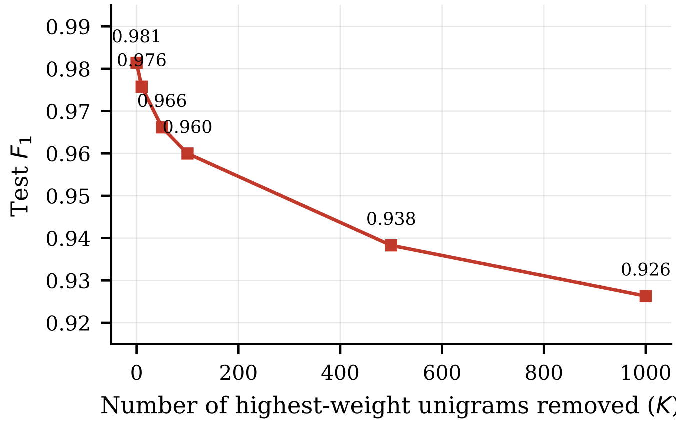

Fig. 5. Shortcut concentration. Deleting the K highest-weight unigrams from the vocabulary degrades F_1 only gradually; at $K = 1000$ the model remains within 5.5 points of the unmodified baseline. The discriminative signal is distributed across many stylistic features rather than concentrated in a few.

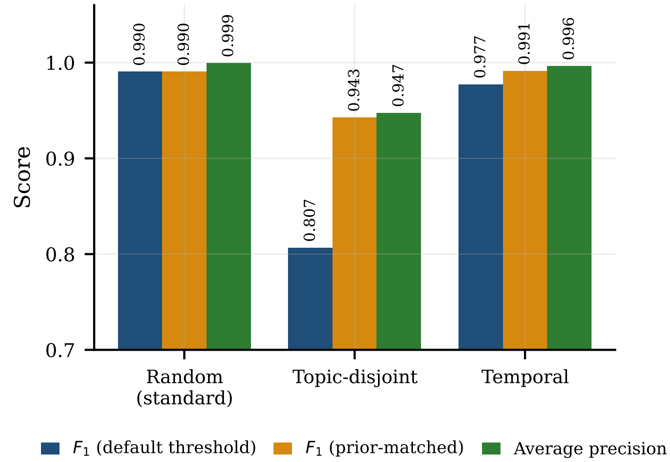

Fig. 7. Selected model across protocols. The gap between the two F_1 bars under topic-disjoint evaluation is the contribution of class-prior shift and threshold miscalibration; the drop in average precision is the genuine loss of discriminative power. Temporal shift leaves both essentially intact.

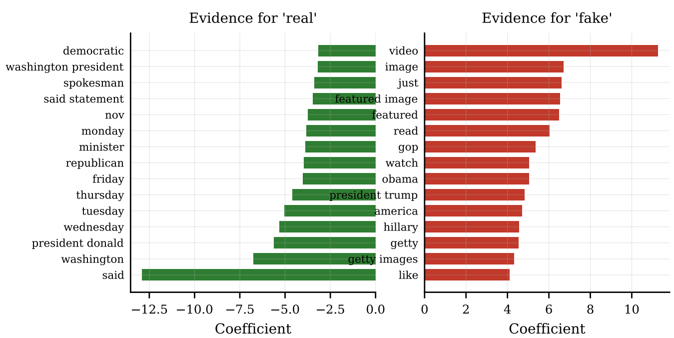

Fig. 6. Highest-magnitude logistic-regression coefficients under the primary configuration. Real-class evidence reflects newswire house style; fake-class evidence reflects web-publishing conventions such as embedded media and image credits. Neither group encodes veracity.

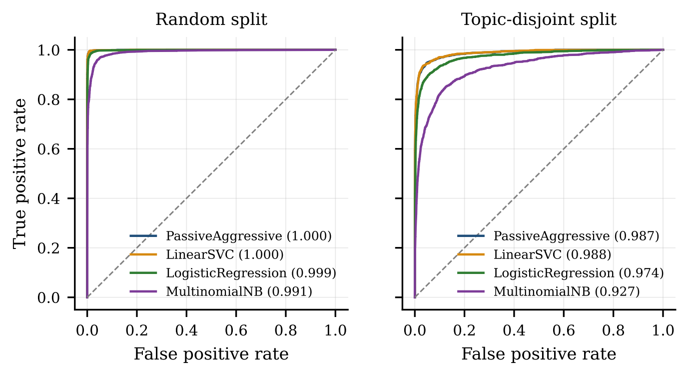

Fig. 8. ROC curves under the random (left) and topic-disjoint (right) protocols, with ROC-AUC in parentheses. ROC-AUC remains high under topic shift even where deployed F_1 collapses, illustrating why threshold-free ranking metrics alone can mask an unusable operating point when the positive class becomes rare.

2017 window, so a chronological split—the protocol often recommended as a realism improvement—does *not* expose the shortcut. Only perturbing provenance and topic does. Practitioners seeking a stress test for this corpus should therefore prefer topic- or source-disjoint splits over temporal ones.

E. Q4: Pretrained Capacity Amplifies the Shortcut

Table VII and Figure 9 report the DistilBERT control against the selected linear model on identical documents and splits.

In-distribution the transformer is clearly the stronger classifier, as expected: F_1 rises from 0.9905 to 0.9993 and average precision from 0.9995 to 1.0000 (to four decimals; the unrounded value is 0.999999). The random-split benchmark therefore rewards capacity, and a study reporting only this row would conclude that a pretrained transformer is the better system.

Under topic shift the ordering reverses sharply, and it does so on precisely the metrics that are robust to the prior. Average precision falls from 1.0000 to 0.8711, a loss of 12.9 points against the linear model's 5.2; prior-matched F_1 falls to 0.6886 ± 0.0027 against the linear model's 0.9427; and balanced accuracy falls to 0.6131 against 0.9475. At its actual operating point the transformer collapses to $F_1 = 0.2592$ with precision 0.1489 at recall 0.9992: it labels almost everything fake, and since the shifted test prior is 0.119, a precision of 0.1489 is only marginally better than a degenerate always-fake rule. Two qualifications keep this honest. The ranking is not destroyed—an average precision of 0.8711 remains far above the 0.119 expected of a random ranker—and the threshold-oracle value of 0.8023 shows that a well-chosen cutoff would recover much of the deployed loss. But every prior-robust comparison places the transformer well below the linear model off-distribution, so the reversal is not an artefact of calibration alone.

Temporal shift again shows the opposite pattern: the transformer holds at $F_1 = 0.9922$ and average precision 0.9999, marginally ahead of the linear model. This reinforces the conclusion of Section VII-D that a chronological split is not a meaningful stress test for this corpus, for either model class.

TABLE VI

PRIOR-CONTROLLED DISTRIBUTION-SHIFT ANALYSIS. F_1^{DEF} IS THE DEPLOYED OPERATING POINT; F_1^{ORC} IS THE BEST F_1 ATTAINABLE BY ANY THRESHOLD ON THE MODEL'S OWN SCORES; AP IS AVERAGE PRECISION; F_1^{MATCH} IS F_1 ON SUBSAMPLES WHOSE CLASS PRIOR IS RESAMPLED TO THAT OF THE RANDOM PROTOCOL (MEAN $\pm$ S.D. OVER M=20 DRAWS). COMPARING F_1^{DEF} ACROSS PROTOCOLS CONFLATES PRIOR SHIFT WITH LOSS OF DISCRIMINATION; AP AND F_1^{MATCH} DO NOT.

Protocol	Model	π_{fake}	P	R	F_1^{def}	F_1^{orc}	AP	F_1^{match}
Random (standard)	Logistic regression	0.461	0.9906	0.9723	0.9814	0.9839	0.9985	0.9814
	LinearSVC	0.461	0.9933	0.9869	0.9901	0.9921	0.9995	0.9901
	Passive–aggressive	0.461	0.9933	0.9877	0.9905	0.9918	0.9995	0.9905
Topic-disjoint	Logistic regression	0.119	0.5596	0.9330	0.6996	0.8372	0.9018	0.9100±.0043
	LinearSVC	0.119	0.7023	0.9460	0.8061	0.8886	0.9482	0.9408±.0036
	Passive–aggressive	0.119	0.7011	0.9498	0.8067	0.8887	0.9475	0.9427±.0032
Temporal	Logistic regression	0.113	0.9098	0.9723	0.9400	0.9577	0.9897	0.9783±.0018
	LinearSVC	0.113	0.9681	0.9867	0.9773	0.9797	0.9963	0.9908±.0011
	Passive–aggressive	0.113	0.9670	0.9880	0.9774	0.9809	0.9964	0.9914±.0013

Truncation does not explain the result. Repeating the random protocol with a 512-token limit changes F_1 only from 0.9993 to 0.9994 (Table VII), so the 256-token window is not the binding constraint on what the model can see.

The most practically alarming number in the table is not a test score but a validation score. Under the topic-disjoint protocol the selected checkpoint had validation $F_1 = 0.9995$—higher than under the random protocol—while its test F_1 was 0.2592. Because the validation split must be drawn from the training pool when no target-domain labels exist, standard model selection returned a confident, well-validated model that fails almost completely on the target distribution. No amount of care with the validation set detects this; only a shifted evaluation does.

Taken together, Q4 answers in the direction that the data, not the instrument, is the binding constraint. Capacity sufficient to represent veracity-bearing content did not cause the model to use it. Instead the transformer fit the provenance signature more sharply—reaching an essentially perfect in-distribution ranking—and paid for it with a steeper loss when that signature stopped being predictive.

F. Computational Cost

Table VIII reports measured cost on the hardware of Section VI. Vectorising 27,177 documents takes 12.19 s and dominates training, which needs 0.44 s for logistic regression and under 0.2 s for the other three; the full four-model pipeline completes in well under a minute on a single CPU. The serialised pipeline occupies 2.15 MB and sustains 3,002 documents per second in batch mode, with 0.66 ms single-document latency. Two consequences follow. First, the audit is inexpensive: the complete linear study, comprising 20 fitted conditions, runs in roughly twelve minutes on one CPU, and the DistilBERT control adds about 32 minutes on one laptop GPU. The diagnostics we recommend therefore impose no practical barrier. Second, on cost grounds the linear pipeline would be an attractive deployment candidate—it needs no accelerator, and it holds the better off-distribution operating point of the two arms—which is precisely why its failure under topic shift matters. Scalability is favourable in the regime that

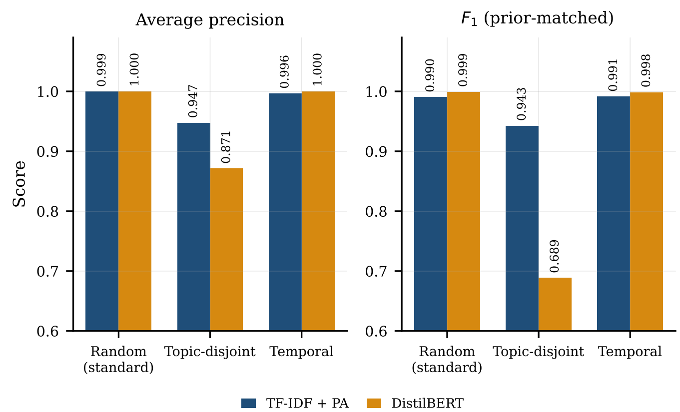

Fig. 9. Prior-robust comparison across protocols. DistilBERT matches or exceeds the linear model in-distribution and under temporal shift, but loses substantially more under topic shift on both average precision and prior-matched F_1. Greater capacity fits the provenance signature more sharply rather than replacing it with veracity-bearing evidence.

matters here: memory grows as $O(Nz)$ rather than $O(NV)$, and inference is independent of N, so throughput would be maintained on substantially larger corpora, with vocabulary construction the first component to require distribution.

VIII. DISCUSSION

A. What the Benchmark Score Measures

Assembling the evidence: the corpus can be solved exactly from a metadata field (Section VII-B); after that field and two further leakage channels are removed, performance falls by only 1.21 points; the surviving signal is insensitive to representation, acquirable from a few hundred examples, and robust to deleting a thousand of its strongest features (Section VII-C); the features it relies on are recognisably editorial conventions (Figure 6); and it degrades measurably when provenance and topic change but not when only time changes (Section VII-D). The consistent explanation is that high scores on this benchmark quantify *source and topic separability*. In the notation of Section III, the pipeline is an efficient estimator of s, and because $I(s; y)$ is near-maximal in $\mathcal{P}_{\text{bench}}$ by the

TABLE VII

CAPACITY CONTROL: FINE-TUNED DISTILBERT AGAINST THE SELECTED LINEAR MODEL ON IDENTICAL DOCUMENTS AND SPLITS. COLUMNS FOLLOW TABLE VI. THE TRANSFORMER WINS IN-DISTRIBUTION ON EVERY METRIC AND LOSES OFF-DISTRIBUTION ON EVERY PRIOR-ROBUST METRIC. THE FINAL ROW REPEATS THE RANDOM PROTOCOL AT A 512-TOKEN LIMIT, CONFIRMING THAT TRUNCATION IS NOT THE OPERATIVE FACTOR. VALIDATION F_1 IS SHOWN TO EXPOSE THE MODEL-SELECTION TRAP: UNDER TOPIC SHIFT IT IS AT ITS HIGHEST EXACTLY WHERE TEST PERFORMANCE IS AT ITS WORST.

Protocol	Model	Val. F_1	P	R	F_1^{def}	F_1^{orc}	AUC	AP	Bal. acc.	F_1^{match}
Random (standard)	TF–IDF + PA	0.9924	0.9933	0.9877	0.9905	0.9918	0.9995	0.9995	0.9910	0.9905
	DistilBERT	0.9992	**0.9997**	**0.9989**	**0.9993**	**0.9997**	**1.0000**	**1.0000**	**0.9993**	**0.9993**
Topic-disjoint	TF–IDF + PA	0.9928	**0.7011**	0.9498	**0.8067**	**0.8887**	**0.9872**	**0.9475**	**0.9475**	$\mathbf{0.9427}\pm.0032$
	DistilBERT	0.9995	0.1489	**0.9992**	0.2592	0.8023	0.9654	0.8711	0.6131	$0.6886\pm.0027$
Temporal	TF–IDF + PA	0.9684	**0.9670**	0.9880	0.9774	0.9809	0.9993	0.9964	0.9918	$0.9914\pm.0013$
	DistilBERT	0.9956	0.9857	**0.9988**	**0.9922**	**0.9946**	**1.0000**	**0.9999**	**0.9985**	**0.9985**
Random, 512 tokens	DistilBERT	0.9997	0.9994	0.9994	0.9994	0.9996	1.0000	1.0000	0.9995	0.9994

TABLE VIII

MEASURED COMPUTATIONAL COST OF THE LINEAR PIPELINE (SINGLE CPU, NO GPU; $N = 27{,}177$ TRAINING DOCUMENTS, $V = 50{,}000$). THE DISTILBERT CONTROL, FOR COMPARISON, REQUIRES 5.8–6.5 MINUTES OF GPU TIME PER PROTOCOL.

Quantity	Value
Vectorisation (fit + transform train)	12.19 s
Training: logistic regression / MNB	0.44 s / 0.016 s
Training: LinearSVC / passive–aggressive	0.16 s / 0.091 s
Non-zeros in training matrix	5.17×10^6
Mean non-zeros per document (z)	190.1 (0.38% density)
Sparse / dense memory for training matrix	$\approx$ 62 MB / $\approx$ 10.9 GB
Serialised pipeline size	2.15 MB
Batch throughput	3,002 docs/s
Single-document latency	0.66 ms

asymmetry of Equation (2), an estimator of s is nearly an estimator of y in-distribution. Nothing in our results indicates that veracity-bearing content c is being used.

This is a claim about the benchmark, not about the feasibility of the task. It also does not imply prior work is invalid: reported numbers are reproducible under the protocols used. What does not follow is the inference from those numbers to deployment readiness.

B. Why Model Capacity Does Not Help

The capacity control of Section VII-E lets us make this argument from our own evidence rather than by inference from the literature's near-ceiling clustering [7], [24]. A pretrained transformer, able in principle to represent the veracity-bearing content c that a bag of words cannot, did not use it. It fit the provenance signature s more sharply instead—achieving an essentially perfect in-distribution ranking—and then lost 2.5 times more average precision than the linear model when s ceased to be predictive. Capacity was spent on the shortcut, not on an alternative to it.

The explanation is that the limitation is an identifiability problem in the data rather than a modelling deficiency. $\mathcal{P}_{\text{bench}}$ contains no counterfactual pairs—same publisher, different veracity—so no objective computed on it can distinguish a rule keyed on c from a rule keyed on s. Both achieve identical empirical risk. Gradient descent then selects whichever is easier to extract, and provenance style is trivially easier than factual verification. A higher-capacity model does not resolve this ambiguity; it merely resolves the easy side of it more precisely, which is why its in-distribution advantage and its out-of-distribution disadvantage appear together. This is the mechanism Schuster et al. [16] describe for machine-generated text, and our metadata-only and deletion probes are the analogue of hypothesis-only diagnostics in NLI [10], [11].

There is a corollary worth stating plainly, because it inverts a common assumption. On a benchmark whose labels are confounded with provenance, a *higher* reported score is weak evidence of a better system and can be evidence of a worse one. Our transformer would be selected over the linear model by any standard random-split comparison, and it is the worse choice for deployment on unseen sources.

C. Metric Choice and Deployment Risk

The shift analysis carries a practical lesson beyond this corpus. Under topic-disjoint evaluation, ROC-AUC remains at 0.9872 for the selected model while deployed F_1 sits at 0.8067 (Figure 8). A study reporting only ROC-AUC would therefore describe a system that is, at its actual operating point, wrong about 30% of its positive predictions. Because misinformation is rare in realistic streams, precision as defined in Equation (10) is the quantity that governs harm—false positives suppress legitimate journalism—and it is exactly the quantity most sensitive to the prior shift that accompanies deployment. We therefore recommend reporting average precision together with the operating point actually used, and treating calibration as a first-class concern rather than a post-hoc adjustment [63]–[65].

D. Recommended Diagnostics

The probes used here are cheap—each is minutes of CPU time—and we suggest them as routine reporting for any study using this corpus or one built by pairing publisher sets:

1) **Metadata-only baseline.** Train on non-textual fields alone. If it solves the task, textual results on the same feature set are uninterpretable.

2) **Small-sample baseline.** Report performance at 1–2% of the training data. Near-ceiling performance there indicates a shallow signal.
3) **Duplicate audit.** Report the fraction of test documents occurring verbatim in training, before de-duplication.
4) **Topic- or source-disjoint split.** Report alongside the random split. Our results indicate this is a far more sensitive stress test than a temporal split for this corpus.
5) **Prior-controlled reporting.** When protocols change the class prior, report average precision and prior-matched F_1 so that calibration effects are not mistaken for lost discrimination.
6) **Do not trust in-domain validation as a shift warning.** Our transformer's validation F_1 was 0.9995—its highest across protocols—in the very condition where test F_1 was 0.2592. A validation split drawn from the training pool, which is all that is available without target labels, provides no signal about this failure. Report a shifted evaluation or report nothing about generalisation.

IX. Limitations

We state the boundaries of these conclusions explicitly.

Single corpus. Every experiment uses one dataset. Our conclusions concern this corpus and, by extension, corpora built by pairing disjoint publisher sets; we do not claim they transfer to benchmarks with different construction, such as claim-level resources [3], [22]. A cross-corpus evaluation is the most important missing experiment and is the first item of Section X.

Instrument capacity. The primary probe is a bag-of-words linear model that cannot represent word order beyond bigrams, negation, or discourse structure. The DistilBERT control (Section VII-E) addresses the obvious objection, but it does not license an unlimited claim. We show that one widely used pretrained model, under a standard fine-tuning recipe, exploits the shortcut more rather than less; we do not show that no model or training procedure could learn veracity-bearing features from this corpus. Our claim remains asymmetric in that direction.

Scope of the capacity control. The transformer result rests on a single architecture, a single seed, two epochs, and no hyperparameter search. Fine-tuning outcomes vary across seeds [57], so the precise magnitudes—particularly the deployed F_1 of 0.2592, which sits at a badly calibrated operating point—should be read as one draw rather than a tight estimate. The *direction* of the effect is supported by three prior-robust metrics simultaneously (average precision, balanced accuracy, prior-matched F_1) and by the 512-token replication, which is why we state the direction confidently and the magnitude loosely. A larger model, a multi-seed protocol, and a search over learning rate and epoch count would each sharpen this.

Asymmetric preprocessing in the control. The transformer receives minimally normalised text while the linear model receives alpha-only text (Section IV-F). Document sets, splits, and leakage controls are identical, but this surface difference means the two arms are not preprocessed identically. It favours the transformer, so it cannot explain the transformer's worse off-distribution result; it does inflate its in-distribution advantage by an amount we have not measured.

GPU nondeterminism. Unlike the linear pipeline, the transformer runs mixed-precision CUDA kernels and is therefore reproducible only to approximately three decimal places.

Confounded shift protocol. Subject values proxy simultaneously for topic, sub-publication, and time-of-collection, so the topic-disjoint protocol perturbs several factors at once. Our prior-matched analysis removes the class-prior confound but not this one; the resulting AP drop should be read as the effect of a bundle of provenance-related shifts rather than of topic alone. A source-level split with publisher metadata—absent from this distribution—would be cleaner.

Exact-duplicate detection only. De-duplication matches identical cleaned strings. Near-duplicates (syndicated rewrites, partial overlaps) are not detected, so the 19.37% contamination figure is a lower bound and the 0.9814 primary score may still be mildly optimistic.

Residual artifacts. We remove one agency name. Other provenance traces (wire-service phrasing, content-management boilerplate) remain and are, by the argument of Section VII-C, precisely what the model uses. We do not claim to have produced an artifact-free version of this corpus; we claim the opposite, namely that doing so by token filtering is not feasible.

English, single period, no external evidence. The corpus is English and concentrated in 2016–2017 US politics, and the pipeline consults no knowledge source, so it cannot in principle verify a claim.

Statistical scope. Bootstrap intervals and McNemar tests quantify uncertainty on a fixed test split; cross-validation quantifies split sensitivity. Neither accounts for variation across corpus construction choices, which our ablations address descriptively rather than inferentially.

X. Future Work

Cross-corpus transfer. The decisive experiment is to train here and evaluate on an independently constructed corpus—NELA-GT [23] for article-level text, LIAR [22] for claim-level—with label mappings stated explicitly. Our prediction, which the present results support but do not establish, is a substantially larger drop than under topic-disjoint evaluation.

Source-controlled data construction. The identifiability problem motivates corpora containing veracity variation *within* publisher, so that s is uninformative about y by design. Retracted or corrected articles from a single outlet are one route; matched-pair sampling on topic and outlet is another.

Scaling the capacity control. Our DistilBERT result (Section VII-E) is a single architecture at a single seed. Extending it to a multi-seed protocol, to full-size encoders, and to instruction-tuned decoder models would establish whether the amplification we observe grows or diminishes with scale—the question that matters for whether current large models are safe to trust on provenance-confounded benchmarks.

Debiasing and invariance. Adversarial removal of source-predictive directions, reweighting, or invariant-risk objectives could be evaluated by whether they improve topic-disjoint AP without degrading in-distribution performance—the trade-off our protocol is designed to expose.

Calibration under shift. Given that most of the deployed F_1 collapse was calibration, methods that maintain calibration without target labels [65] deserve evaluation in this setting.

Beyond exact duplicates. Near-duplicate detection (shingling, MinHash) would tighten the contamination estimate and yield a cleaner corpus version.

XI. Conclusion

We audited a widely used fake news corpus by holding a transparent TF–IDF and linear-classifier pipeline fixed and varying the data and the evaluation protocol. The pipeline reproduces the literature's near-ceiling result ($F_1 = 0.9905$, 95% CI $[0.9883, 0.9928]$), and we then established what that number rests on. A metadata-only classifier that never reads the article text solves the benchmark exactly ($F_1 = 1.000$). Removing that field, a newswire source tag present in 99.21% of real articles, and duplicates that contaminate 19.37% of a naive test split costs only 1.21 F_1 points, because the residual signal is diffuse editorial style: nine representation variants span 0.62 points, 543 documents recover 95.1% of full-data performance, and deleting the $1{,}000$ highest-weight unigrams leaves F_1 at 0.9263. That signal does not transfer. Under a topic-disjoint protocol average precision falls from 0.9995 to 0.9475 and deployed F_1 to 0.8067; a prior-matched analysis attributes most of the latter to class-prior shift and miscalibration (0.9427) while confirming a genuine 5.2-point loss of discrimination. Temporal transfer is by contrast nearly lossless, which identifies topic- and source-disjoint splits—not chronological ones—as the informative stress test for this corpus.

A capacity control settles where the limitation lies. Fine-tuned DistilBERT, trained on identical documents and splits, is the stronger classifier in-distribution ($F_1 = 0.9993$, average precision 1.0000) and the weaker one under topic shift on every prior-robust metric, losing 12.9 points of average precision against the linear model's 5.2 and reaching prior-matched F_1 of 0.6886 against 0.9427. Capacity sufficient to represent veracity-bearing content was instead spent fitting the provenance signature more sharply. Notably, its validation F_1 peaked at 0.9995 in precisely the condition where test F_1 fell to 0.2592, so standard in-domain model selection gives no warning of the failure.

Our contribution is measurement rather than modelling, and its limits are correspondingly specific: one corpus, a shift protocol that perturbs several provenance factors together, and a capacity control at one architecture and one seed (Section IX). Within those limits the practical implication is concrete. Reported accuracy on this benchmark should be read as a measurement of source and topic separability, and studies using it should accompany their headline number with the metadata-only, small-sample, duplicate-audit, and topic-disjoint diagnostics of Section VIII, each of which costs minutes of CPU time. For practitioners, the operative finding is that a system scoring 0.99 in-distribution can mispredict roughly 30% of its positive decisions once the topic mix changes—a failure mode invisible to threshold-free metrics and to random-split evaluation alike. All code, seeds, and derived results are released to make each of these checks a matter of re-execution rather than reimplementation.

Appendix A
Hyperparameters

Table IX lists every setting required to reproduce the reported numbers. Values are not tuned: defaults were fixed before the experiments to avoid an implicit search that the single fixed test split could not support. A consequence is that reported figures are not upper bounds; the ablations of Table V show the achievable range is narrow in any case.

TABLE IX
Complete hyperparameter listing.

Component	Parameter	Value
TF–IDF	`ngram_range`	$(1, 2)$
	`min_df`	5
	`max_df`	0.9
	`max_features`	50,000
	`sublinear_tf`	true
	`stop_words`	English (built-in list)
	`strip_accents`	unicode
Logistic regression	C, solver, `max_iter`	1.0, liblinear, 1000
Multinomial NB	α	0.1
LinearSVC	C, loss	1.0, squared hinge
Passive–aggressive	C, `max_iter`	1.0, 1000
DistilBERT	checkpoint	`distilbert-base-uncased`
	max sequence length	256 (512 in the sensitivity run)
	batch size, epochs	16, 2
	learning rate, warmup	2×10^{-5}, 10% linear
	precision	fp16 autocast
	selection	best per-epoch validation F_1
Protocol	split	$70/10/20$, stratified
	seed σ	42
	min. document length	20 characters
	selection metric	validation F_1
Statistics	bootstrap resamples	2000
	prior-match repeats M	20

Appendix B
Learning-Curve and Deletion-Probe Values

Numerical values underlying Figure 4 (logistic-regression probe, primary configuration, random protocol), as training documents $\rightarrow$ test F_1: $135 \rightarrow 0.6802$; $271 \rightarrow 0.8704$; $543 \rightarrow 0.9334$; $1{,}358 \rightarrow 0.9370$; $2{,}717 \rightarrow 0.9579$; $6{,}794 \rightarrow 0.9714$; $13{,}588 \rightarrow 0.9765$; $27{,}177 \rightarrow 0.9814$.

Values underlying Figure 5, as unigrams removed $K \rightarrow$ test F_1: $0 \rightarrow 0.9814$; $10 \rightarrow 0.9758$; $50 \rightarrow 0.9662$; $100 \rightarrow 0.9600$; $500 \rightarrow 0.9383$; $1000 \rightarrow 0.9263$. The twenty highest-weight unigrams, removed first, are: *said, video, washington, image, just, featured, read, gop, wednesday, watch, tuesday, obama, america, thursday, hillary, getty, like, friday, republican, minister*.

Appendix C
Corpus Composition

Figure 10 shows class counts in the raw distribution and the token-length distribution after normalisation. The length distributions overlap substantially, so length alone is a weak cue; the bimodality visible in the real class reflects the mixture of short wire briefs and full reports.

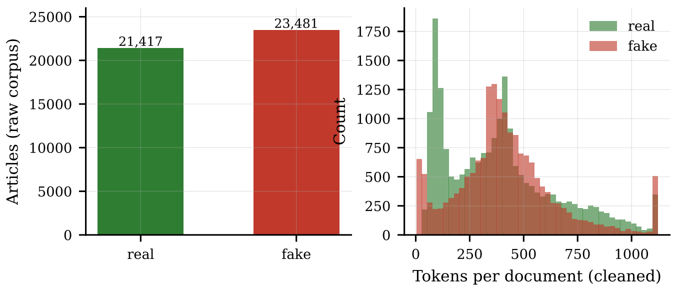

Fig. 10. Raw class counts (left) and token-length distributions after normalisation, clipped at the 98th percentile for legibility (right).

Appendix D
Reproducibility Checklist

- **Code.** Released at https://github.com/vermayuvraj/fake-news-detection, including the audit scripts that regenerate every table and figure.
- **Data.** Not redistributed; obtained from [8]. Construction from the raw files is fully specified by Algorithm 1.
- **Environment.** Exact versions pinned (Section VI); no GPU required.
- **Randomness.** Single seed $\sigma = 42$ for splits and models; deterministic solver and frequency-ordered vocabulary. Two independent runs of the linear pipeline produced bit-identical metrics; the DistilBERT control is seeded identically but, running mixed-precision CUDA kernels, reproduces to approximately three decimal places.
- **Split alignment.** The transformer script asserts that its split cardinalities equal those of the linear runs before training, so the capacity comparison cannot silently drift out of alignment.
- **Protocol integrity.** Model selection on validation only; the test split is evaluated once per condition; the vectoriser is fitted on training data only.
- **Negative control.** Label-permutation run yields test ROC-AUC $= 0.4974$.
- **Uncertainty.** Bootstrap 95% intervals (B=2000), five-fold cross-validation, McNemar exact tests, and M=20 prior-matched resamples.
- **Artifacts.** Machine-readable results (`results.json`, `shift_analysis.json`, `transformer.json`) accompany the code, and all reported values are drawn from them.
- **Compute.** Linear study $\approx$ 12 minutes on one CPU; transformer control $\approx$ 32 minutes on one RTX 4050 (three protocols at 6.5, 5.8 and 6.2 minutes, plus 13.0 minutes for the 512-token replication).

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